

13-Machine_Learning_Pipeline

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1 Machine Learning Pipeline

```
[96]: import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.impute import SimpleImputer
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import make_pipeline, Pipeline
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer

from sklearn.tree import DecisionTreeClassifier

import joblib
```

```
[97]: d1 = {
    'Social_media_followers' : [1000000, np.nan, 2000000, 1310000, 1700000, np.
    ↪nan, 4100000, 1600000, 2200000, 1000000],
    'Sold_out': [1,0,0,1,0,0,0,1,0,1]
}

d2 = {
    'Genre':['Rock', 'Metal', 'Bluegrass', 'Rock', np.nan, 'Rock', 'Rock',
    ↪np.nan, 'Bluegrass', 'Rock'],
    'Social_media_followers':[1000000, np.nan, 2000000, 1310000, 1700000,
    ↪np.nan, 4100000, 1600000, 2200000, 1000000],
    'Sold_out':[1,0,0,1,0,0,0,1,0,1]
}

df1 = pd.DataFrame(d1)
df1
```

```
[97]:   Social_media_followers  Sold_out
0          1000000.0          1
1              NaN          0
2          2000000.0          0
3          1310000.0          1
```

4	1700000.0	0
5	NaN	0
6	4100000.0	0
7	1600000.0	1
8	2200000.0	0
9	1000000.0	1

```
[98]: X1 = df1[['Social_media_followers']]
      y1 = df1[['Sold_out']]

      X1_train, X1_test, y1_train, y1_test = train_test_split(X1, y1, test_size=0.3,
      ↪random_state=19)

      imputer = SimpleImputer(strategy='mean')
      lr = LogisticRegression()

      pipe1 = make_pipeline(imputer, lr)

      pipe1.fit(X1_train, y1_train)
```

c:\Users\ikiga\AppData\Local\Programs\Python\Python311\Lib\site-packages\sklearn\utils\validation.py:1229: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().

```
y = column_or_1d(y, warn=True)
```

```
[98]: Pipeline(steps=[('simpleimputer', SimpleImputer()),
                       ('logisticregression', LogisticRegression())])
```

```
[99]: pipe1.score(X1_train, y1_train)
```

```
[99]: 1.0
```

```
[100]: pipe1.score(X1_test, y1_test)
```

```
[100]: 0.6666666666666666
```

```
[101]: pipe1.named_steps.simpleimputer.statistics_
```

```
[101]: array([2051666.66666667])
```

```
[102]: pipe1.named_steps.logisticregression.coef_
```

```
[102]: array([[ -9.72872687e-05]])
```

1.0.1 More Advance Pipeline

```
[103]: df = pd.DataFrame(data=d2)
df
```

```
[103]:
```

	Genre	Social_media_followers	Sold_out
0	Rock	1000000.0	1
1	Metal	NaN	0
2	Bluegrass	2000000.0	0
3	Rock	1310000.0	1
4	NaN	1700000.0	0
5	Rock	NaN	0
6	Rock	4100000.0	0
7	NaN	1600000.0	1
8	Bluegrass	2200000.0	0
9	Rock	1000000.0	1

```
[104]: X = df.iloc[:,0:2]
y = df.iloc[:,2]

X_train, X_test, y_train, y_test = train_test_split(X,y, test_size=0.3,
↳random_state=17)

num_cols = ["Social_media_followers"]
cat_cols = ['Genre']

num_pipeline = Pipeline(
    steps = [
        ('impute', SimpleImputer(strategy='mean')),
        ('scale', StandardScaler())
    ]
)

cat_pipeline = Pipeline(steps=[
    ('impute', SimpleImputer(strategy='most_frequent')),
    ('one-hot-encoder', OneHotEncoder(handle_unknown='ignore',
↳sparse_output=False))
])
cat_pipeline
```

```
[104]: Pipeline(steps=[('impute', SimpleImputer(strategy='most_frequent')),
                        ('one-hot-encoder',
                         OneHotEncoder(handle_unknown='ignore', sparse_output=False))])
```

```
[105]: col_transformer = ColumnTransformer(transformers= [
    ('num_pipeline', num_pipeline, num_cols),
    ('cat_pipeline', cat_pipeline, cat_cols),
],
```

```
remainder='drop', n_jobs=-1
)
```

```
[106]: dtc = DecisionTreeClassifier()
pipefinal = make_pipeline(col_transformer, dtc)
pipefinal.fit(X_train, y_train)
```

```
[106]: Pipeline(steps=[('columntransformer',
                        ColumnTransformer(n_jobs=-1,
                                          transformers=[('num_pipeline',
                                                          Pipeline(steps=[('impute',
                                                                              SimpleImputer()),
                                                                              ('scale',
                                                                              StandardScaler()))],
                                                                              ['Social_media_followers']),
                                                                              ('cat_pipeline',
                                                                              Pipeline(steps=[('impute',
                                                                              SimpleImputer(strategy='most_frequent')),
                                                                              ('one-hot-
                                                                              encoder',
                                                                              OneHotEncoder(handle_unknown='ignore',
                                                                              sparse_output=False))]),
                                                                              ['Genre'])])),
                        ('decisiontreeclassifier', DecisionTreeClassifier())])
```

```
[107]: pipefinal.score(X_test, y_test)
```

```
[107]: 0.6666666666666666
```

1.1 How to save your pipeline

```
[108]: joblib.dump(pipefinal, 'pipe.joblib')
```

```
[108]: ['pipe.joblib']
```

```
[109]: pipefinal2 = joblib.load('pipe.joblib')
pipefinal2
```

```
[109]: Pipeline(steps=[('columntransformer',
                        ColumnTransformer(n_jobs=-1,
                                          transformers=[('num_pipeline',
                                                          Pipeline(steps=[('impute',
                                                                              SimpleImputer()),
                                                                              ('scale',
                                                                              StandardScaler()))],
                                                                              ['Social_media_followers']),
                                                                              ('cat_pipeline',
```

```

SimpleImputer(strategy='most_frequent')),
                                                    Pipeline(steps=[('impute',
                                                                    ('one-hot-
encoder',
OneHotEncoder(handle_unknown='ignore',
sparse_output=False)))]),
                                                    ['Genre']]])),
('decisiontreeclassifier', DecisionTreeClassifier())])

```